

PANKAJ KUMAR JANGID

Current position: PhD Research Scholar, IISER Pune India
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SUMMARY

PhD researcher in theoretical chemical physics working at the interface of stochastic processes, nonequilibrium statistical mechanics, chemical reaction networks, and stochastic thermodynamics. My research develops physically interpretable mathematical models for catalytic reactions, molecular transport, and complex chemical systems using stochastic thermodynamics and statistical mechanics. I am interested in understanding energy transduction, self-organization, and nonequilibrium phenomena in reaction networks, while integrating scientific machine learning to accelerate theoretical discovery.

RESEARCH INTEREST

Nonequilibrium Statistical Mechanics, Stochastic Thermodynamics, Chemical Reaction Networks, Nonequilibrium Transport, Statistical Physics of Complex Systems, Physics-Informed Machine Learning.

RESEARCH EXPERIENCE

2021–Present **Ph.D. Researcher**, Indian Institute of Science Education and Research (IISER) Pune

- Developed theoretical stochastic and analytical models of chemical reaction networks and catalytic processes, showing how microscopic fluctuations govern energy transduction and catalytic efficiency.
- Developed mathematical frameworks based on statistical mechanics, stochastic processes, and reaction kinetics to explain nonequilibrium phenomena in catalytic and biological systems.
- Modeled Markov jump and non-Markovian processes to study first-passage phenomena, dynamic catalysis, and anomalous transport in complex environments.
- Developed C++/Python frameworks for large-scale stochastic simulation and numerical validation of theoretical models.

2020–2021 **Master's Thesis Researcher**, Indian Institute of Science Education and Research (IISER), Mohali

- Developed numerical methods to study quantum resonance states, and analyzed resonance tunneling through phase-space distributions.

EDUCATION

2021 - present PhD (Chemical Physics) at **Indian Institute of Science Education and Research (IISER) Pune**

Thesis: Stochastic Dynamics and Nonequilibrium Phenomena in Complex Biological and Catalytic Systems

2016 - 2021 BS-MS Degree in Chemistry at **Indian Institute of Science Education and Research (IISER) Mohali**

Thesis: Revisiting Siegert Pseudostates Method for Calculation of Resonances

SKILLS

Analytical & Computational Modeling	Chemical Reaction Networks, Stochastic Processes, Chemical Master Equation, Brownian Dynamics, Monte Carlo Simulation, Non-Markovian Dynamics
Scientific Computing	Numerical Optimization, High-Performance Computing (HPC)
Mathematics	Probability Theory, Statistics, Statistical Mechanics, Differential Equations
Programming	Python (NumPy, SciPy, pandas, Matplotlib, PyTorch), C++, MATLAB, Mathematica
Machine Learning	PyTorch, Deep Learning, Physics-Informed Neural Networks, Transformers, Diffusion Models

SCIENTIFIC TALKS AND POSTERS

- 2025 Delivered a scientific talk at *Chemsymphoria 2025*, IISER Pune, Pune, India.
- 2025 Poster presentation at *Inter IISER–NISER Chemistry Meet*, IISER Pune, India.
- 2024 Poster presentation at *Society of Physical Chemistry (SoPhyC) Conference*, Indian Institute of Technology Bombay, Mumbai, India.
- 2024 Poster presentation at *RSC Meet the Editors*, IISER Pune & CSIR–National Chemical Laboratory, Pune, India.
- 2024 Poster presentation at *International Conference on Fundamental and Advanced Research in Chemistry*, Indian Institute of Technology Mandi, India.
- 2023 Poster presentation at *Theoretical Chemistry Symposium*, Indian Institute of Technology Madras, Chennai, India.

PUBLICATIONS

Research Highlights: 10 peer-reviewed publications (9 first-author); one additional first-author manuscript under revision.

1. **Jangid, P.;** Chaudhury, S. *Anomalous diffusion of coupled active–passive particles in viscoelastic media*. Under peer review (2026).
2. **Jangid, P.;** Chaudhury, S.; Kolomeisky, A. *Dynamic Fluctuations Can Counteract Catalytic Inhibition and Enhance Catalytic Performance*. *ACS Catal.* **2026** 16, 16403–16415.
3. **Jangid, P.;** Chaudhury, S. *DNA target search in chromatin compartments under stochastic resetting*. *J. Phys. Chem. B* **2026**, 130, 3766–3775.
4. **Jangid, P.;** Chaudhury, S. *Stochastic dynamics of nanoparticle catalysis: A discrete-state perspective*. *Mater. Horiz.* **2026**, 13, 2152–2172.
5. **Jangid, P.;** Metzler, R.; Chaudhury, S. *Internal protein motion in a rough model potential*. *J. Chem. Phys.* **2025**, 163, 244101.
6. **Jangid, P.;** Chaudhury, S.; Kolomeisky, A. *Effect of temperature in dynamic catalysis*. *J. Phys. Chem. C* **2025**, 129, 15606–15618.
7. **Jangid, P.;** Chaudhury, S.; Kolomeisky, A. *Theoretical understanding of dynamic catalysis*. *J. Phys. Chem. C* **2024**, 128, 9077–9089.
8. **Jangid, P.;** Chaudhury, S. *Transition Path Dynamics of Non-Markovian Systems across a Rough Potential Barrier*. *J. Phys. Chem. A* **2024**, 128, 10041–10052.
9. **Jangid, P.;** Punia, B.; Chaudhury, S. *Stochastic dynamics of hairballs in single-polymer growth*. *Phys. Chem. Chem. Phys.* **2024**, 26, 29749–29758.

10. Chaudhury, S.; **Jangid, P.**; Kolomeisky, A. B. *Dynamics of chemical reactions on single nanocatalysts with heterogeneous active sites. J. Chem. Phys.* **2023**, 158, 074101.
11. **Jangid, P.**; Chaudhury, S. *Exploring the role of heterogeneity in quorum sensing cells: A discrete-state stochastic approach. J. Stat. Mech.* **2023**, 2023, 093501.

AWARDS & FELLOWSHIPS

- 2026 ANRF-ITS Travel Grant, Anusandhan National Research Foundation (ANRF), India
- 2024 Best Poster Award, Royal Society of Chemistry – CrystEngComm Symposium, IISER Pune
- 2023 Best Teaching Assistant Award, Department of Chemistry, IISER Pune
- 2023 CSIR-UGC NET, Council of Scientific and Industrial Research (CSIR-UGC), India
- 2021 PhD Research Fellowship, IISER Pune
- 2021 Graduate Aptitude Test in Engineering (GATE), Indian Institute of Technology (IIT) Bombay, India
- 2020 Certificate of Merit, IISER Mohali
- 2016 Joint Entrance Examination (JEE Advanced), Indian Institute of Technology (IIT) Guwahati, India

TEACHING

Worked as a Teaching Assistant (TA) in *Quantum Chemistry, Organic Chemistry, Thermodynamics* at IISER Pune, India.

PROFESSIONAL SERVICE

- 2024 Organizer, ChemSymphoria2024, IISER Pune. Managed conference finances and event logistics.

SELECTED MACHINE LEARNING PROJECTS

Physics-Informed Neural Networks for Non-Markovian Diffusion

 [GitHub](#)

- Developed a physics-informed neural network (PyTorch) to solve memory-dependent diffusion equations with temporal convolution kernels.
- Learned both the concentration field and unknown memory kernel directly from sparse and noisy observations.
- Implemented the complete framework from scratch, including automatic differentiation, quadrature-based memory integrals, custom loss functions, and training pipeline.

Anomalous Hurst Coefficient Prediction using Deep Sequence Models

 [GitHub](#)

- Developed deep learning models for estimating the Hurst exponent of fractional Brownian motion trajectories.
- Implemented simulation pipelines (Davies–Harte, Hosking, Cholesky), feature-based baselines, Bidirectional LSTM, and Transformer regressors.
- Evaluated models using MAE, RMSE, and R^2 metrics with comprehensive visualization and analysis tools.